

Fundamental Theorem of Galois Theory

Vincent Wang

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1 Introduction

P 18.19. (a).

$$A = \begin{pmatrix} \sigma_1(\alpha_1) & \sigma_2(\alpha_1) & \sigma_3(\alpha_1) & \dots & \sigma_n(\alpha_1) \\ \sigma_1(\alpha_2) & \sigma_2(\alpha_2) & \sigma_3(\alpha_2) & \dots & \sigma_n(\alpha_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_1(\alpha_m) & \sigma_2(\alpha_m) & \sigma_3(\alpha_m) & \dots & \sigma_n(\alpha_m) \end{pmatrix}.$$

Let X denote the column of basis vectors $(\alpha_1, \alpha_2, \dots, \alpha_n)$. Denote by $\sigma(X) = (\sigma(\alpha_1), \sigma(\alpha_2), \dots, \sigma(\alpha_n))$. The columns of A are of the form $\sigma_j(X)$. Suppose FTSOC the columns of A are not linearly independent over K^n . That is, there exist $a_1, a_2, \dots, a_n \in K$ such that $\sum_{j=1}^n a_j \sigma_j(X) = \mathbf{0}$. Now define the function $\tau : K \rightarrow K$ such that $\tau = \sum_{j=1}^n a_j \sigma_j$. The α_i span K , so for any $x \in K$ there exist $b_1, b_2, \dots, b_n \in k$ such that $x = \sum_{j=1}^n b_j \alpha_j$. Now

$$\begin{aligned} \tau(x) &= \sum_{j=1}^n a_j \sigma_j(x) \\ &= \sum_{j=1}^n a_j \sum_{i=1}^n \sigma_j(b_i) \sigma_j(\alpha_i) \\ &= \sum_{j=1}^n \sum_{i=1}^n a_j b_i \sigma_j(\alpha_i) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_j b_i \sigma_j(\alpha_i) \\ &= \sum_{i=1}^n b_i \sum_{j=1}^n a_j \sigma_j(\alpha_i) \\ &= \sum_{j=1}^n 0 \\ &= 0. \end{aligned}$$